Vaibhav Saxena

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Summary

Software Engineer with 2 years of experience designing, building, and shipping production systems end-to-end. Strong computer science fundamentals (Data Structures & Algorithms, OOP, OS, Computer Networks, Multithreading & Concurrency) with a focus on clean, secure, high-performance code; also builds GenAI / LLM systems (MCP, RAG, LangGraph, LLM evaluation).

Experience

ADROSONIC Jan 2024 – Present
Software Engineer (Research and Innovation Engineer) *Mumbai, India*

- Delivered load-balancer autoscaling: built a pipeline that syncs HAProxy backends with VM scale groups via Ansible AWX, with reconciliation so no new VM misses traffic.
- Redesigned platform-wide authentication after a VAPT flagged 365-day tokens, shipping 30-min access / rotating refresh tokens on Keycloak (OAuth 2.0 + PKCE) with stateless JWT validation at the gateway, cutting per-request auth DB queries **75%** (4 → 1) and adding fingerprint-bound cookies to block token replay.
- Re-engineered the end-to-end customer onboarding pipeline for **30,000+ users**, integrating Stripe and Razorpay payment gateways and revamping the signup flow to improve onboarding conversion by **15%**.
- Led development of an LLM-assisted entity-resolution engine (PySpark, Delta Lake, Microsoft Fabric) that deduplicates records with no shared keys; multi-key blocking cut pairwise comparisons by **99%** (86K → 547) and batched, cached LLM adjudication resolved **157 ambiguous pairs in 3 API calls**, with union-find clustering.
- Led development of an event-driven schema-drift service that detects renamed, new and retyped columns via sampled-value type inference and LLM-scored matching to a canonical schema, auto-resolving confident renames and reusing stored reviewer decisions for repeat drift.

ADROSONIC Jun 2023 – Dec 2023
Software Engineer Intern (Research and Innovation Engineer Intern) *Mumbai, India*

- Built a config-driven data-quality framework (PySpark, Microsoft Fabric) evaluating 15 rules over **8M+ rows** in a single pass per table, cutting runtime by **87%** (2 hrs → 15 min) and surfacing **250K+ duplicate** and **1K+ orphaned** source records, each routed to a named owner via Power BI dashboards and Teams alerts.
- Replaced Stripe's hosted Checkout with a custom payment flow built on the Stripe Payment Intents & Payment Methods APIs, enabling saved cards for returning users and removing repeated billing-address entry to improve checkout conversion.

Projects

Limit Order Book, FIX Engine & Agentic Simulation | GitHub | *C++, Python, MCP, Lock-Free Concurrency*

- Built a multithreaded C++ matching engine with price-time priority, fed by a lock-free MPSC ring buffer (atomic CAS, acquire/release ordering, `alignas(64)` padding vs. false sharing), with $O(1)$ cancels via `unordered_map` + `std::list`.
- Implemented a FIX 4.2 encode/decode layer (`NewOrderSingle`, `ExecutionReport`) and exposed the engine to Python through an `extern "C"` library and `ctypes`.
- Wrapped the engine in an MCP server driving a multi-agent market simulation with market maker, momentum, spoofer and detector agents.

ClauseIQ | GitHub · Live Demo | *Python, LangGraph, MCP, FastAPI, ChromaDB, GCP*

- Built and deployed a multi-agent system (LangGraph, Python) that flags unfair clauses in Indian contracts and cites the governing section of law, served through a FastAPI + SSE API and an MCP server callable from Claude Desktop.
- Engineered hybrid retrieval (BM25 + dense embeddings, Reciprocal Rank Fusion) over the Indian Contract Act in ChromaDB, reaching **1.00 citation accuracy** and **0.90 faithfulness** at about **$0.007/contract**, validated by a CI-gated DeepEval (LLM-as-judge) harness on a 20-contract benchmark.
- Deployed on GCP Cloud Run (Gemini via Vertex AI, scale-to-zero) with a Vercel frontend, shipped through keyless GitHub Actions CI/CD using WIF.

Technical Skills

Languages: C++ (11/14/17), Python, SQL

Backend & APIs: FastAPI, Django REST Framework, Node.js, REST APIs

AI & Agentic: Model Context Protocol (MCP), LangGraph, RAG & Hybrid Retrieval, LLM Integration, Agentic Workflows, Tool Calling, DeepEval (LLM evaluation)

Cloud & DevOps: Microsoft Azure, Fabric, GCP (Cloud Run, Vertex AI), AWS (Lambda, EventBridge, S3, Bedrock, Textract, Comprehend Medical), Terraform, Docker, CI/CD (GitHub Actions), Git

Data & Databases: PySpark, MySQL, Redis

Frontend: React, Angular, TypeScript

Achievements

- **LeetCode Knight**: max rating 1857, best global rank 1186.
- **National Finalist, Google GenAI Exchange Hackathon**: one of India's top teams, presented at the finals in Bengaluru.
- **G.P. Birla Merit Scholarship**: awarded for top-percentile academic performance during B.Tech at BIT Mesra.

Education

Birla Institute of Technology, Mesra 2020 – 2024
B.Tech, CGPA: 8.53 / 10 *Ranchi, Jharkhand*